A MORPHISM OF SIMPLICIAL SETS THAT IS BIJECTIVE BELOW N, SURJECTIVE AT N AND COSKELETAL ABOVE IS A TRIVIAL KAN FIBRATION

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Editorial note. The counted statement is the author's own lemma environment labelled lemma-section (source lines 6198-6211); the proof environment begins two lines below it and is its own proof, closing at source line 6238.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; horns, skeleta and coskeleta are the author's surrounding development in the same chapter.

Editorial note. This card lies in the half of the chapter assigned to this lane under the line-number split (statement start >= 3601); the candidate was selected with that filter before reading.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Objects chapter, counted once: a morphism that is bijective below n, surjective at n and coskeletal above is a trivial Kan fibration.

Allowed standard prerequisites (editorial).

- Simplicial set, horn and Kan fibration. The Kan condition (every horn extends) and the notion of a trivial Kan fibration, which additionally induces surjections in every degree.. It is the conclusion and the shape of what has to be verified (source lines 6210, 6213-6220).
- Skeleton and coskeleton. The n-skeleton and n-coskeleton functors on simplicial sets and the canonical morphisms from an object to the coskeleton of its skeleton.. The hypotheses (3) and (4) are the isomorphisms with the coskeleton, used in the high-dimensional case (source lines 6205-6208, 6225-6237).
- Standard simplex and its boundary. The standard k-simplex with its boundary, and the fact that for k > n the n-skeletons of the two coincide.. It is what makes the extension automatic for horns of dimension greater than n (source lines 6230-6237).

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Definition 0.1. A map $X \to Y$ of simplicial sets is called a *trivial Kan fibration* if $X_0 \to Y_0$ is surjective and for all $n \geq 1$ and any commutative solid diagram

$$\begin{array}{ccc} \partial\Delta[n] & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow \\ \Delta[n] & \longrightarrow & Y \end{array}$$

a dotted arrow exists making the diagram commute.

Lemma 0.2. *Let $f : V \to U$ be a morphism of simplicial sets. Let $n \geq 0$ be an integer. Assume*

(1) *The map $f_i : V_i \to U_i$ is a bijection for $i < n$.*
(2) *The map $f_n : V_n \to U_n$ is a surjection.*
(3) *The canonical morphism $U \to cosk_n sk_n U$ is an isomorphism.*
(4) *The canonical morphism $V \to cosk_n sk_n V$ is an isomorphism.*

Then f is a trivial Kan fibration.

Proof. Consider a solid diagram

$$\begin{array}{ccc} \partial\Delta[k] & \longrightarrow & V \\ \downarrow & \nearrow & \downarrow \\ \Delta[k] & \longrightarrow & U \end{array}$$

as in Definition 0.1. Let $x \in U_k$ be the k-simplex corresponding to the lower horizontal arrow. If $k \leq n$ then the dotted arrow is the one corresponding to a lift $y \in V_k$ of x; the diagram will commute as the other nondegenerate simplices of $\Delta[k]$ are in degrees $< k$ where f is an isomorphism. If $k > n$, then by conditions (3) and (4) we have (using adjointness of skeleton and coskeleton functors)

$$\mathrm{Mor}(\Delta[k], U) = \mathrm{Mor}(\mathrm{sk}_n\Delta[k], \mathrm{sk}_n U) = \mathrm{Mor}(\mathrm{sk}_n\partial\Delta[k], \mathrm{sk}_n U) = \mathrm{Mor}(\partial\Delta[k], U)$$

and similarly for V because $\mathrm{sk}_n\Delta[k] = \mathrm{sk}_n\partial\Delta[k]$ for $k > n$. Thus we obtain a unique dotted arrow fitting into the diagram in this case also. □